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Calculating the CMB power spectrum from simulated cosmic structure formation initiated by primordial fluctuations

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ABSTRACT

I will write an abstract for the whole project at the end, once I have completed all the milestones.

Key words. cosmic microwave background – large-scale structure of Universe – linear perturbations

0. Introduction

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1. Background cosmology

To compute the CMB power spectrum, we must first establish the background cosmological model upon which perturbations will evolve. This section details the construction of this background Universe. We begin by outlining the theoretical framework, most notably the Friedmann-Lemaître-Robertson-Walker (FLRW) metric and the resulting Friedmann equations that governs the expansion history and the evolution of its energy components. Subsequently, we describe the numerical implementation of this model, including the fiducial parameters adopted from Planck 2018. Finally, we validate our implementation by comparing its predicted timeline of cosmic epochs and derived quantities, such as the age of the universe, against theoretical expectations, and we present the results of an MCMC parameter fit to supernova data, which contextualizes the well-known Hubble tension within our model.

1.1. Theory

In this section, we will present the relevant theory for developping the background cosmology in our model Universe along with all of the equations used. All of the theory is covered in greater detail in the lecture notes by Hans A. Winther for the cosmology course AST5220 (Winther 2026a,b), so we will simply summarise the key details and derive some quantities that will come in handy for the numerical implementation.

1.1.1. The Friedmann-Lemaître-Robertson-Walker geometry

For our background cosmology, we assume an isotropic and homogeneous Universe with a spatial scale factor $a(t)$ equipped with the Friedmann-Lemaître-Robertson-Walker (FLRW) metric

$$ds^2 = -dt^2 + a^2(t)\Big(dx^2 + dy^2 + dz^2\Big), \tag{1}$$

or, on component form; $g_{00} = -1$, $g_{ij} = a^2(t)\delta_{ij}$.

At the background level, the constituents of our Universe, which include radiation (photons and massless neutrinos), non-relativistic matter (baryons and cold dark matter), and a cosmological constant, are treated as perfect fluids. For each component i, the stress-energy tensor in the comoving frame takes the diagonal form

$$T^{(i)}_{\mu\nu} = \mathrm{diag}(\rho_i, p_i, p_i, p_i),$$

with ρ_i denoting the density and p_i the pressure of that component.

The resulting Einstein field equations for the FLRW metric and the sum of the stress-energy tensors for the perfect fluids,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \sum_i T^{(i)}_{\mu\nu},$$

we get the Friedmann equations. The (00) component of the Einstein equations gives us the first equation,

$$H^2(a) \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\sum_i \rho_i, \tag{2}$$

while the (11), (22) and (33) components gives us the second Friedmann equation (assuming isotropy, or else they would yield three different equations),

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\sum_i (\rho_i + 3p_i). \tag{3}$$

As our time variable, we will use the logarithm of scale factor $x \equiv \ln[a(t)]$. We thus need to study the Friedmann equations, Eqs.(2)-(3), to understand how the scale factor evolves and how it is related to the evolution of the densities of the different components in our model Universe.

* The code developped for this project is available at GitHub.

1.1.2. The Hubble function

Introducing the definition of the density parameters $\Omega_i \equiv 8\pi G\rho_i/3H^2$, and the equation of state $w_i \equiv p_i/\rho_i$, the first Friedmann equation can be recast into the form we will be using for our implementation of the background cosmology

$$H^2 = H_0^2 \sum_i \frac{\Omega_{i0}}{a^{3(1+w_i)}},$$

where $H_0 = 100h$ km/s/Mpc is the Hubble constant, and Ω_{i0} is the current day value of the density parameter of component i.

In our model Universe, we have dark energy, described by a cosmological constant (Λ) with equation of state $w_\Lambda = -1$, non-relativistic matter in the form of baryons (b) and cold dark matter (CDM) with equation of state $w_M = 0$, and two relativistic species, photons (γ) and neutrinos (ν) with $w_R = 1/3$. The curvature (k) can be decribed as having an equation of state $w_k = -1/3$. We can thus explicitly write out our Hubble function as

$$H(a) = H_0 \sqrt{\Omega_{\Lambda 0} + \frac{\Omega_{k0}}{a^2} + \frac{\Omega_{b0} + \Omega_{\rm CDM0}}{a^3} + \frac{\Omega_{\gamma 0} + \Omega_{\nu 0}}{a^4}}. \quad (4)$$

The evolution of the density parameters as functions of the scale factor are thus given by

$$\Omega_k(a) = \frac{\Omega_{K0}}{a^2 H(a)^2/H_0^2}, \qquad \Omega_M(a) = \frac{\Omega_{M0}}{a^3 H(a)^2/H_0^2},$$
$$\Omega_R(a) = \frac{\Omega_{R0}}{a^4 H(a)^2/H_0^2}, \qquad \Omega_\Lambda(a) = \frac{\Omega_{\Lambda 0}}{H(a)^2/H_0^2}. \quad (5)$$

Here we define the non-relativistic matter density parameter $\Omega_M = \Omega_b + \Omega_{\rm CDM}$ and the relativistic species density parameter $\Omega_R = \Omega_\gamma + \Omega_\nu$.

1.1.3. Cosmic and conformal time

In the FLRW metric, Eq.(1), we use the *cosmic time* coordinate t which is related to the scale factor through the Hubble function through the relation

$$H \equiv \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t} \;\Rightarrow\; \frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{H},$$

or on integral form

$$t(x) = \int_{-\infty}^{x} \frac{\mathrm{d}x'}{H(x')}. \quad (6)$$

However, we also consider the *conformal time* η which is the distance that light may have travelled since the Big Bang (horizon). It is related to the cosmic time thorugh the relation

$$a\mathrm{d}\eta = c\mathrm{d}t \;\Rightarrow\; \frac{\mathrm{d}\eta}{\mathrm{d}x} = \frac{c}{aH}.$$

For calculating η, it is also prudent to introduce the conformal Hubble function $\mathcal{H} \equiv \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}\eta} = aH$ such that the conformal time can be written on integral form as

$$\eta(x) = \int_{-\infty}^{x} \frac{c\mathrm{d}x'}{\mathcal{H}(x')}. \quad (7)$$

The derivatives of $\mathcal{H}$ with respect to x are given by;

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} = \mathcal{H}\left[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\right]\Bigg|_{a=e^x}, \quad (8)$$

$$\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} = \mathcal{H}\Bigg\{\left[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\right]^2 - \frac{1}{2}\left(3\Omega_M + 4\Omega_R + 2\Omega_K\right)^2 + \frac{1}{2}\left(9\Omega_M + 16\Omega_R + 4\Omega_K\right)\Bigg\}\Bigg|_{a=e^x}, \quad (9)$$

where Ω_i are the density parameters evaluated at $a = e^x$. The derivatives of $\mathcal{H}$ are derived in Appendix A.

1.1.4. Luminosity distance and angular diameter distance

In order to compare our model with observations, and be able to perform parameter fitting, we will need to calculate two observables; namely the *angular diameter distance* d_A, and the *luminosity distance* d_L.

Writing the line element in the FLRW metric, Eq.(1), using spherical coordinates, we have

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2\left(\frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\theta^2 + r^2\sin\theta^2\mathrm{d}\phi^2\right), \quad (10)$$

where k is the curvature. Since we are observing photons (that move radially along the line of sight) when determining astrophysical distances, we consider the null geodesic, $\mathrm{d}s^2 = 0$, followed by the rays of light. For these photons, we get

$$c\mathrm{d}t = \frac{a\mathrm{d}r}{\sqrt{1-kr}} \;\Rightarrow\; \chi \equiv \eta_0 - \eta = \int_0^r \frac{\mathrm{d}r'}{\sqrt{1-kr'}}. \quad (11)$$

Here, χ denotes the *comoving distance*.

Solving the integral in Eq.(11) and isolating r, we find three solutions, depending on the curvature

$$r = \begin{cases} \dfrac{\chi \sin\left[\sqrt{|\Omega_{k0}|}H_0\chi/c\right]}{\sqrt{|\Omega_{k0}|}H_0\chi/c} & \Omega_{k0} < 0, \\ \chi & \Omega_{k0} = 0, \\ \dfrac{\chi \sinh\left[\sqrt{|\Omega_{k0}|}H_0\chi/c\right]}{\sqrt{|\Omega_{k0}|}H_0\chi/c} & \Omega_{k0} > 0. \end{cases}$$

If an object has physical size D and subtends an angle θ on the sky, the angular diameter distance is defined as $d_A = D/\theta$. Reading off the line element, Eq.(10), a displacement in the θ-direction corresponds to a proper distance $dD = ar\,d\theta$, so

$$d_A = D/\theta = ar.$$

On the other hand, from the intrinsic luminosity of a source and its observed flux, the luminosity distance d_L can be defined through $F = L/4\pi d_L^2$, which simply reads

$$d_L = a/r = d_A/a^2.$$

1.1.5. Some cosmological events

Finally, for our analysis of the simulated data, it is convenient to identify key cosmological events to serve as heuristic markers for our interpretation of the results. We will look at three relevant events at the time scale of the evolution of the background:

(i) matter-radiation equality at $x_{\rm eq}^{MR} = \ln\left(\frac{\Omega_{R0}}{\Omega_{M0}}\right)$;

(ii) the onset of cosmic acceleration (when $\ddot{a} = 0$) at

$$x_{\rm acc} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{2\Omega_{\Lambda 0}}\right);$$

(iii) and the matter-dark energy equality at

$$x_{\rm eq}^{\Lambda M} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{\Omega_{\Lambda 0}}\right).$$

For the derivations of the three x's, see Appendix B.

1.2. Implementation details

1.2.1. Fiducial cosmological parameters

Our model Universe will assume ΛCDM fiducial cosmology and be initialised with the best-fit cosmological parameters from the Planck 2018 data (Aghanim et al. 2020). We will thus take the dimensionless Hubble parameter $h = 0.67$, the current day temperature of the CMB radiation $T_{\rm CMB0} = 2.7255$ K, and the effective number of massless neutrino species $N_{\rm eff} = 3.046$.

For the current day values of the density parameters, we set

$$\begin{aligned}
\Omega_{b0} &= 0.05,\\
\Omega_{\rm CDM0} &= 0.267,\\
\Omega_{K0} &= 0,\\
\Omega_{\gamma 0} &= \frac{\pi^2}{15}\frac{(k_b T_{\rm CMB0})^4}{\hbar^3 c^5}\frac{8\pi G}{3H_0^2} \simeq 5.509\times 10^{-5},\\
\Omega_{\nu 0} &= \frac{7}{8}\left(\frac{4}{11}\right)^{4/3} N_{\rm eff}\Omega_{\gamma 0} \simeq 3.811\times 10^{-5},\\
\Omega_{\Lambda 0} &= 1 - \sum_{i\neq\Lambda}\Omega_{i0} \simeq 0.683.
\end{aligned}$$

Here, the dark energy density parameter is defined such that $\sum_i \Omega_{i0} = 1$, to ensure a spatially flat Universe.

1.2.2. Numerical integration scheme

To evaluate integral quantities such as Eqs.(6) and (7), we recast them as ordinary differential equations and integrate numerically using the fourth-order Runge-Kutta solver provided by the GSL library (GNU Project 2024) and spline the result using the cubic spline algorithm available in GSL.

The cosmic time ODE is

$$\frac{{\rm d}t}{{\rm d}x} = \frac{1}{H(x)},$$

which is to be integrated up from $x_i = -\infty$ to some final x_f. However, in practice, we must choose a very early time to integrate from. In our case, we solve in the interval $x_i = -20$ up to $x_f = 5$. The initial condition for the ODE is determined deep inside the radiation dominated era, when $H(x) \sim H_0\sqrt{\Omega_{R0}}e^{-2x}$, such that

$$\frac{{\rm d}t}{{\rm d}x} = \frac{e^{2x}}{H_0\sqrt{\Omega_{R0}}} \;\Rightarrow\; t(x_i) = \int_{-\infty}^{x_i}\frac{e^{2x}{\rm d}x}{H_0\sqrt{\Omega_{R0}}} = \frac{1}{2H(x_i)}.$$

Similarly, we solve the ODE for the conformal time

$$\frac{{\rm d}\eta}{{\rm d}x} = \frac{c}{\mathcal{H}(x)}$$

in the same interval, with the initial condition $\eta(x_i) = c/\mathcal{H}(x_i)$, due to the radiation dominated conformal Hubble function $\mathcal{H}(x) \sim H_0\sqrt{\Omega_{R0}}e^{-x}$.

1.2.3. MCMC parameter fitting

Using Markov Chain Monte Carlo (MCMC), we will test our simulated Universe, based on fiducial cosmological parameters, against observations of type Ia supernova (SNIa) luminosity distances (Betoule et al. 2014). The procedure searches for the triplet of "best-fit" parameters in the $(h, \Omega_{M0}, \Omega_{K0})$ parameter space that minimises

$$\chi^2(h,\Omega_{M0},\Omega_{K0}) = \sum_{i=1}^{N}\frac{[d_L(z_i,h,\Omega_{M0},\Omega_{K0}) - d_L^{\rm obs}(z_i)]^2}{\sigma_i^2}.$$

Here, d_L are the predicted luminosity distances from our model at the observed redshifts z_i, while $d_L^{\rm obs}$ are the observed distances, together with their associated errors.

To perform the actual MCMC parameter fitting procedure, we employ the code written by Hans A. Winther (Winther 2026c).

1.3. Results

1.3.1. Timeline of the background

In Table 1, we have stated the times for the matter-radiation equality, the onset of cosmic expansion, and the matter-dark energy equality, in terms of the log scale factor x, the redshift $z(x) = e^{-x} - 1$ and the cosmic time $t(x)$. We remark that cosmic acceleration only begins in the very late Universe at around $z \sim 0.6$. We thus observe the same late-stage acceleration as we would expect from the ΛCDM Universe.

Displayed in Figure 1 are the density parameters Ω_R, Ω_M, and Ω_Λ as functions of x, representing the contributions from relativistic components, non-relativistic matter, and the cosmological constant. The figure highlights three distinct periods in the history of our model Universe:

(i) an era of radiation domination in the very early Universe until $x_{\rm eq}^{MR} = -8.132$;

(ii) a matter dominated period between $x_{\rm eq}^{MR} = -8.132$ and $x_{\rm eq}^{\Lambda M} = -0.256$;

(iii) and a late stage dark energy dominated Universe after $x_{\rm eq}^{\Lambda M} = -0.256$.

This timeline is explained by the fact that $\rho_R \propto a^{-4}$, $\rho_M \propto a^{-3}$, while $\rho_\Lambda = $ const, such that the density of the relativistic components decays faster than non-relativistic matter, which in turn decays faster than the density of dark energy.

Equality times and onset of acceleration			
Event	Log scale factor x	Redshift z	Time t [Gyr]
Matter-radiation equality (MR)	-8.132	3400.33	5.106×10^{-5}
Onset of acceleration (acc)	-0.487	0.6272	7.752
Matter-dark energy equality (ΛM)	-0.256	0.2915	10.378

Table 1. Times for when we have matter-radiation equality, onset of cosmic acceleration, and matter-dark energy equality expressed using x, redshift $z = e^{-x} - 1$, and cosmic time $t(x)$.

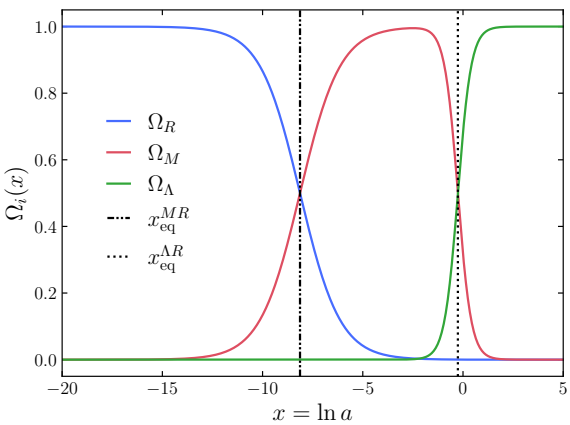


Fig. 1. Evolution of the density parameters of relativistic species (Ω_R), non-relativistic matter (Ω_M), and of dark energy (Ω_Λ) as functions of $x = \ln a$ on the interval $-20 \leq x \leq 5$. Plotted with vertical dashed and dotted lines are also the times of matter-radiation and matter-dark energy equality.

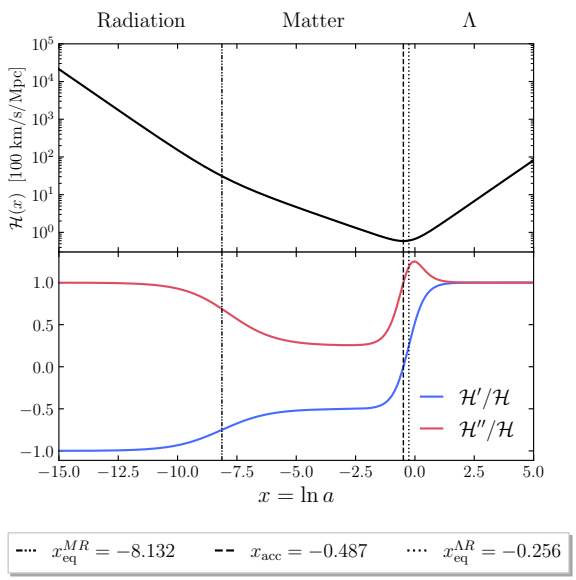


Fig. 2. The conformal Hubble function $\mathcal{H}(x)$ along with $\mathcal{H}'/\mathcal{H}$ and $\mathcal{H}''/\mathcal{H}$ plotted as functions of x. The time at which we have the onset of cosmic acceleration $x_{\rm acc}$, and the equality times $x_{\rm eq}^{mr}$ and $x_{\rm eq}^{\Lambda m}$ are shown with vertical dotted lines with the dominating component indicated above the regions in between them.

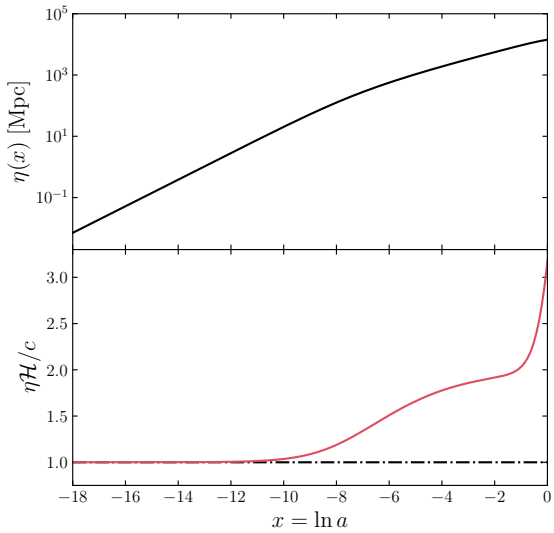


Fig. 3. The top panel presents the evolution of the conformal time with respect to x. The bottom panel shows the dimensionless quantity $\eta\mathcal{H}/c$ over the same interval.

The transition between cosmological eras leaves clear signatures in the conformal Hubble function and its derivatives, as illustrated in Figure 2. Equations (8) and (9) predict specific values in each regime: $\mathcal{H}'/\mathcal{H} = -1$ and $\mathcal{H}''/\mathcal{H} = 1$ in radiation domination, $\mathcal{H}'/\mathcal{H} = -1/2$ and $\mathcal{H}''/\mathcal{H} = 1/4$ in matter domination, and both ratios approaching unity in a Λ-dominated universe. These predicted features are apparent in the bottom panel of Figure 2.

The top panel also exhibits the cosmic acceleration after $x_{\rm acc}$. Since $\mathcal{H} \equiv aH = \dot{a}$, the conformal Hubble function describes the rate of change of the scale factor with respect to cosmic time. We observe that the slope of $\mathcal{H}$ changes from negative to positive, i.e. $\mathcal{H}'/\mathcal{H}$ becomes positive, at the minimum situated at $x = x_{acc}$. A positive slope means that

$$\mathcal{H}' = \frac{\mathrm{d}\dot{a}}{\mathrm{d}x} = \frac{\mathrm{d}t}{\mathrm{d}x}\ddot{a} > 0.$$

Since $t(x)$ is manifestly a strictly increasing function, $\mathrm{d}t/\mathrm{d}x$ is strictly positive at all times, such that the sign of $\mathcal{H}'$ depends on the sign of $\ddot{a}$. Therefore, we see that $\ddot{a} > 0$ for $x > x_{\rm acc}$, so we indeed have late-stage cosmic acceleration.

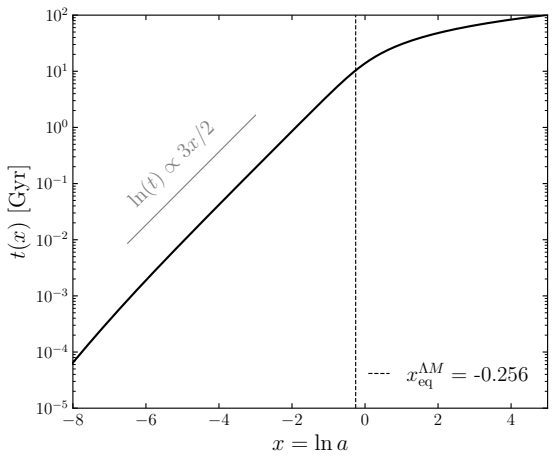


Fig. 4. Cosmic time with respect to x. The gray diagonal line is included in order to compare the slope of $t(x)$ with our expectation of $\ln t \propto 3x/2$ during matter domination.

1.3.2. Cosmic and conformal time, and the age of the Universe

Figure 3 presents the conformal time $\eta(x)$ as well as the dimensionless quantity $\eta\mathcal{H}/c$ in the interval $-18 \leq x \leq 0$. From the initial condition $\eta(x_i) = c/\mathcal{H}(x_i)$, derived in Section 1.2.2, we expect that $\eta\mathcal{H}/c \to 1$ as $x \to -\infty$. We observe that this is in fact the case for our model (see top panel, Fig.3). The conformal time is also a smooth monotonically increasing function of x, and its current day value is calculated to be $\eta(0)/c \simeq 46.319$ Gyr.

The solution for the cosmic time $t(x)$ is showcased in Figure 4 over the interval $-8 \leq x \leq 5$, with a current day value determined to be $t_0 \equiv t(0) \simeq 13.858$ Gyr. During the matter domination era, the Hubble function is $H(x) \propto e^{-3x/2}$, so we would expect that

$$\frac{\mathrm{d}t}{\mathrm{d}x} \propto e^{3x/2} \;\Rightarrow\; \ln t(x) \propto \frac{3}{2}x.$$

Comparing to the gray line included in Figure 4, we see that it matches the slope of $t(x)$ between $x = -8$ up to $x \sim x_{\rm eq}^{\Lambda M}$. In the dark energy dominated era ($x \gtrsim x_{\rm eq}^{\Lambda M}$), the slope decreases and the curve flattens out. This is to be expected from the fact that $H \sim H_0\sqrt{\Omega_{\Lambda 0}} = \text{const}$ such that $t \propto x$ during this era.

1.3.3. MCMC best-fit parameters and the Hubble tension

The MCMC parameter fitting to SNIa luminosity distance data (see Sect.1.2.3) yielded the best-fit triplet of parameters

$$\begin{aligned} h &= 0.701711, \\ \Omega_{M0} &= 0.255027, \\ \Omega_{K0} &= 0.0789514, \end{aligned}$$

with an associated $\chi^2 = 29.2799$. The observational data contains $N = 31$ data, so we have $\chi^2/N \lesssim 1$, indicating a good fit.

Figure 5 shows the $(\Omega_{M0}, \Omega_{\Lambda 0})$ parameter space sampled during the MCMC procedure, highlighting the regions within 1σ and 2σ of the best-fit parameters (minimum χ^2). The fiducial Planck values lie within the 1σ contour, though slightly offset from the best-fit point.

The marginalized posterior distributions for $\Omega_{\Lambda 0}$ and the Hubble parameter h are presented in Figure 6, with vertical dotted lines indicating the fiducial Planck values. While the fiducial

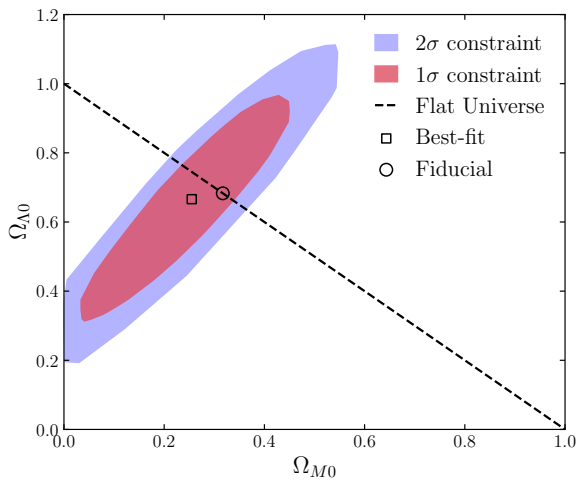


Fig. 5. The regions of sampled points in the $(\Omega_{M0}, \Omega_{\Lambda 0})$ parameter space within 1σ and 2σ of the pair associated with the minimal χ^2. The diagonal dashed line shows the cosmologies giving a flat Universe; $\Omega_{K0} = 1 - \Omega_{M0} - \Omega_{\Lambda 0} = 0$.

$\Omega_{\Lambda 0}$ falls comfortably within the posterior and close to its best-fit value, the same is not true for h. The fiducial value of h lies well outside the 1σ region of its posterior distribution, with a relative difference of

$$\left|\frac{h_{\text{best-fit}} - h_{\text{fiducial}}}{h_{\text{best-fit}}}\right| \simeq 4.5\%.$$

This discrepancy is further illustrated in Figure 7, which compares the luminosity distance-redshift relation predicted by our best-fit model to that of the fiducial Planck cosmology, alongside the observed SNIa data. The best-fit parameters clearly provide a superior match to the supernova measurements across the full redshift range, reinforcing that while the Planck cosmology remains our adopted baseline for subsequent calculations, it is less consistent with these late-universe distance indicators.

Together, these results highlight the well-known *Hubble tension*: the persistent disagreement between the value of H_0 inferred from early-universe observations (such as the CMB) and that derived from late-universe distance measurements (e.g., supernovae or galaxy surveys). Despite this tension, we will adopt the fiducial Planck ΛCDM cosmology for the remainder of this work, while noting its inconsistency with the SNIa luminosity distances used in our MCMC analysis.

2. Recombination history

Having established the background cosmology, we now turn to the recombination history of the Universe. While the background evolution governs the large-scale expansion, it is the microphysics of the photon-baryon plasma that determines when and how the CMB photons we observe today were released. Three epochs are of particular importance: *recombination*, when free electrons and protons combine to form neutral hydrogen at $z \sim 1100$; *decoupling*, when the photon mean free path exceeds the Hubble horizon and the CMB photons stream freely for the first time; and *reionisation*, when the first generation of stars reionises the intergalactic medium at $z \sim 8$. Together, these epochs shape the optical depth of the Universe and leave distinct imprints on the CMB power spectrum. In this section we compute

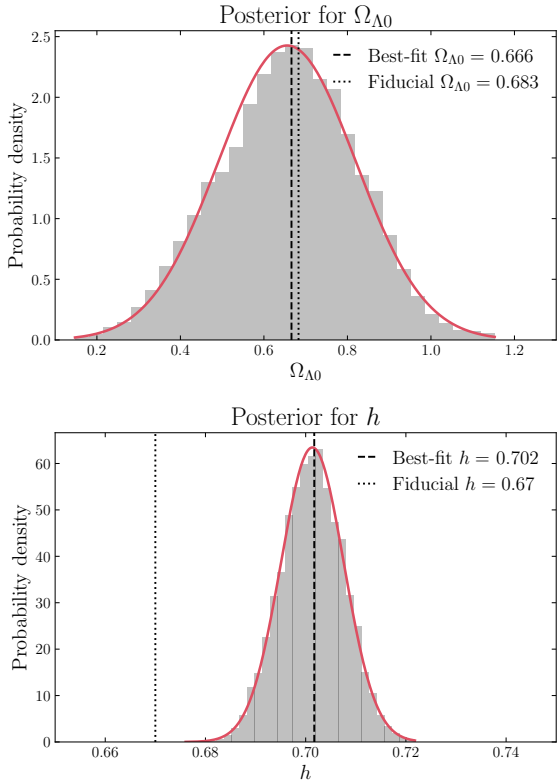


Fig. 6. Posteriors for $\Omega_{\Lambda 0}$ and h from the MCMC parameter fitting. In the posterior for h, we have also indicated the fiducial value of $h = 0.67$ with a blue vertical line to highlight the Hubble tension.

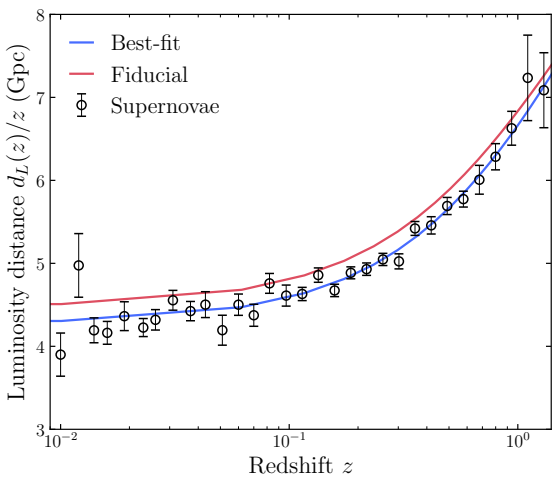


Fig. 7. Luminosity distance as a function of redshift z. The data points represent the luminosity distances deduced from observations of supernovae at different redshifts. The two curves show the luminosity distance functions $d_L(z)/z$ for the best fit parameters and for the fiducial cosmological parameters.

the free electron fraction X_e, the optical depth τ, and the visibility function $\tilde{g}$, which together encode the full scattering history of CMB photons. The theory and methodology follow Winther (2026a,d).

2.1. Theory

At early times, the Universe was so hot that it was fully ionised, with photons and baryons tightly coupled through Thomson scattering in a photon-baryon plasma. As the Universe expanded and cooled, free electrons and protons combined to form neutral hydrogen in a process known as *recombination*:

$$e^- + p^+ \rightleftharpoons H + \gamma.$$

Before recombination, the plasma was opaque to photons, which scattered repeatedly off free electrons and could not travel freely. After recombination, the photon mean free path grew to exceed the Hubble horizon, and the photons decoupled from the baryons, i.e. the Universe became transparent. The photons released at this moment of decoupling form the CMB we observe today, making the recombination history central to understanding the origin and properties of the CMB.

The opacity of a medium to radiation is quantified by the *optical depth* τ. A beam of light with initial intensity I_0 traversing a medium of optical depth τ emerges with intensity $I = I_0 e^{-\tau}$, so that $\tau \ll 1$ corresponds to a transparent medium and $\tau \gg 1$ to an opaque one. In cosmology, the dominant source of opacity is Thomson scattering of photons off free electrons, giving

$$\tau(\eta) = \int_{\eta}^{\eta_0} n_e \sigma_T a \, \mathrm{d}\eta',$$

where n_e is the free electron number density, σ_T is the Thomson cross-section[1], and a is the scale factor. The epoch of last scattering, from which the CMB photons we observe today last scattered, corresponds to $\tau \approx 1$.

In order to calculate the optical depth, we must therefore solve for the electron number density n_e, or in practice its proxy, the free electron fraction $X_e \equiv n_e/n_H$, where n_H corresponds to the number density of protons (hydrogen nuclei).

2.1.1. Electron fraction and number density

In the early Universe, when ionising radiation is still efficient and the plasma remains in thermal equilibrium, the free electron fraction can be determined from the *Saha equation*. As a first approximation, we neglect helium and assume all baryons are protons ($Y_p = 0$), so that the proton number density is

$$n_H \approx n_b = \frac{\rho_b}{m_H} = \frac{3H_0^2\Omega_{b0}}{8\pi G m_H a^3}.$$

Under this assumption, the Saha equation takes the form

$$\frac{X_e^2}{1 - X_e} = \frac{1}{n_b}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\epsilon_0/k_B T_b}, \tag{12}$$

where m_e is the electron mass, k_B is Boltzmann's constant, $\epsilon_0 = 13.6$ eV is the ionisation energy of hydrogen, $\hbar$ is the reduced Planck constant, and T_b is the baryon temperature. In principle, T_b and the photon temperature T_γ should be evolved separately, but since Compton scattering keeps the two tightly coupled until well after recombination, it is an excellent approximation to set $T_b \simeq T_\gamma = T_{\mathrm{CMB},0}/a$ (Winther 2026d, see appendix).

[1] The Thomson scattering cross-section is given by $\sigma_T = \frac{8\pi}{3}\frac{\alpha^2\hbar^2}{m_e^2 c^2}$.

Including helium ($Y_p \neq 0$) complicates the picture, as helium can be both singly and doubly ionised, contributing additional free electrons. We define the helium mass fraction as

$$Y_p \equiv \frac{m_{\rm He} n_{\rm He}}{m_b n_b} = \frac{4 n_{\rm He}}{n_b},$$

where we have approximated the baryon mass as $m_b \approx m_H$, the hydrogen atom mass, and used the fact that each helium nucleus contains four baryons. It follows that the hydrogen number density is

$$n_H = (1 - Y_p)\, n_b = (1 - Y_p) \frac{3 H_0^2 \Omega_{b0}}{8\pi G m_H a^3}.$$

To account for the three ionisation states of hydrogen and helium, we introduce the ionisation fractions

$$u \equiv \frac{n_{\rm He^+}}{n_{\rm He}}, \qquad v \equiv \frac{n_{\rm He^{2+}}}{n_{\rm He}}, \qquad w \equiv \frac{n_{\rm H^+}}{n_{\rm H}},$$

each of which satisfies its own set of three Saha equations

$$\frac{u}{1-u-v} = \frac{2}{n_e}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_0/k_B T_b}, \tag{13}$$

$$\frac{v}{u} = \frac{4}{n_e}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_1/k_B T_b}, \tag{14}$$

$$\frac{w}{1-w} = \frac{1}{n_e}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\epsilon_0/k_B T_b}. \tag{15}$$

Here $\chi_0 = 24.587$ eV is the ionisation energy of neutral helium and $\chi_1 = 4\epsilon_0 = 54.418$ eV is the ionisation energy of He^+. By tallying the free electrons contributed by each ion species, we can define the quantity

$$f_e \equiv \frac{n_e}{n_b} = (2v + u)\,\frac{Y_p}{4} + w\,(1 - Y_p), \tag{16}$$

from which the free electron fraction follows as $X_e = f_e/(1-Y_p)$. The factor of two in $2v$ reflects the fact that each doubly ionised helium nucleus contributes two free electrons.

The Saha approximation is valid as long as the plasma remains in thermal equilibrium, which holds well for $X_e \gtrsim 0.99$. As recombination proceeds and the Universe departs from equilibrium, the Saha equation overestimates the recombination rate and a non-equilibrium treatment is required. The *Peebles equation* (Peebles 1968) provides such a treatment, tracking the non-equilibrium evolution of X_e through the ODE

$$\frac{\mathrm{d}X_e}{\mathrm{d}x} = \frac{C_r(T_b)}{H}\left[\beta(T_b)(1 - X_e) - n_H \alpha^{(2)}(T_b) X_e^2\right].$$

The two terms in brackets represent the competing processes of photoionisation (first term), which increases X_e, and recombination (second term), which decreases it. The prefactor C_r accounts for the fact that recombinations directly to the ground state produce an ionising photon that immediately re-ionises a nearby hydrogen atom, making such recombinations ineffective. Only recombinations to excited states, followed by either the two-photon $2s \to 1s$ decay or escape of a Lyman-α photon, result in a net increase of neutral hydrogen. The various quantities appearing in the Peebles equation are

$$\begin{aligned}
C_r(T_b) &= \frac{\Lambda_{2s\to1s} + \Lambda_\alpha}{\Lambda_{2s\to1s} + \Lambda_\alpha + \beta^{(2)}(T_b)},\\
\Lambda_{2s\to1s} &= 8.227\ \mathrm{s}^{-1},\\
\Lambda_\alpha &= \frac{H}{(8\pi)^2 n_{1s}}\left(\frac{3\epsilon_0}{\hbar c}\right)^3,\\
n_{1s} &= (1 - X_e) n_H,\\
\beta^{(2)}(T_b) &= \beta(T_b)\, e^{3\epsilon_0/4k_B T_b},\\
\beta(T_b) &= \alpha^{(2)}(T_b)\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\epsilon_0/k_B T_b},\\
\alpha^{(2)}(T_b) &= \frac{8\sigma_T c}{\sqrt{3\pi}}\sqrt{\frac{\epsilon_0}{k_B T_b}}\,\phi_2(T_b),\\
\phi_2(T_b) &= 0.448 \ln(\epsilon_0/k_B T_b)\,.
\end{aligned}$$

Here $\Lambda_{2s\to1s}$ is the decay rate of the $2s$ state via two-photon emission, Λ_α is the Lyman-α escape rate, $\alpha^{(2)}$ is the recombination coefficient to excited states ($n \geq 2$), β is the corresponding photoionisation rate, and $\beta^{(2)}$ is the photoionisation rate from the $2s$ state.

2.1.2. Reionisation

Around a redshift of $z \sim 10$, the first stars and galaxies form and begin emitting large amounts of high-energy ultraviolet radiation. This radiation is energetic enough to ionise the surrounding neutral hydrogen and helium, driving the free electron fraction X_e back up from its post-recombination value of $X_e \sim 10^{-4}$ toward values close to unity. This process is known as *reionisation*. Our primary interest is its effect on CMB photons propagating from the surface of last scattering to us today. For this purpose a simple phenomenological model suffices. We will thus use the same simplified model used in CAMB Lewis et al. (2000), adding two tanh-shaped transitions to the Peebles solution

$$\begin{aligned}
X_e = X_e^{\rm Peebles} &+ \frac{1 + f_{\rm He}}{2}\left(1 + \tanh\frac{y_{\rm reion} - y}{\Delta y_{\rm reion}}\right)\\
&+ \frac{f_{\rm He}}{2}\left(1 + \tanh\frac{z_{\rm Hereion} - z}{\Delta z_{\rm Hereion}}\right).
\end{aligned} \tag{17}$$

The first term accounts for the reionisation of hydrogen and the second for the reionisation of helium. Each transition is centred on its respective reionisation redshift and smoothed over a width Δz. Here $y \equiv (1+z)^{3/2} = e^{-3x/2}$ is a rescaled redshift variable, $f_{\rm He} \equiv Y_p/[4(1 - Y_p)]$ is the helium-to-hydrogen number ratio, $y_{\rm reion} = (1 + z_{\rm reion})^{3/2}$, and $\Delta y_{\rm reion} = \frac{3}{2}\sqrt{1 + z_{\rm reion}}\,\Delta z_{\rm reion}$ is the corresponding width in y-space. From the assumed Planck cosmology, we have $z_{\rm reion} = 8$, $\Delta z_{\rm reion} = 0.5$, $z_{\rm Hereion} = 3.5$, and $\Delta z_{\rm Hereion} = 0.5$.

2.1.3. Visibility function

After we have solved for the optical depth τ, we will also compute the *visibility function* $\tilde{g}$ given by

$$\tilde{g}(x) = -\tau'(x) e^{-\tau(x)}. \tag{18}$$

It describes the probability density for a given photon to last time having scattered at time x, and has the property $\int_{-\infty}^{0} \tilde{g}(x)\mathrm{d}x = 1$. The visibility function peaks when the photons decoupled from the baryons, i.e. at the transition between the opaque and transparent epochs (taken to be when $\tau = 1$).

2.1.4. Sound horizon

The final quantity of interest is the *sound horizon* at decoupling which is the total comoving distance a sound wave in the photon-baryon plasma of the early Universe could travel from the Big Bang until photons decouple. This scale length is the maximum distance between regions that could have had causal contact and will thus leave a distinct imprint on both the CMB power spectrum and the large-scale distribution of matter in the Universe. The sound speed of the tightly coupled photon-baryon plasma is somewhat lower than the relativistic sound speed of a pure photon gas ($c/\sqrt{3}$), and is given by

$$c_s = \frac{c\sqrt{R}}{\sqrt{3(1+R)}}, \qquad R \equiv \frac{4\Omega_{\gamma 0}}{3\Omega_{b0}a}.$$

The sound horizon is then

$$s(x) = \int_{-\infty}^{x} \frac{c_s}{\mathcal{H}}\,\mathrm{d}x', \quad \Rightarrow \quad \frac{\mathrm{d}s}{\mathrm{d}x} = \frac{c_s}{\mathcal{H}(x)}, \tag{19}$$

and its value at decoupling $s(x_{\rm dec})$ sets the physical scale of the acoustic peaks in the CMB.

2.2. Implementation details

2.2.1. Solving for the free electron fraction

As outlined in Section 2.1.1, the evolution of the free electron fraction X_e is divided into three distinct epochs, each requiring a different treatment; the equilibrium Saha regime at early times, the out-of-equilibrium Peebles regime during and after recombination, and the reionisation epoch at late times. The calculation of X_e is therefore done in three phases:

(i) first we calculate X_e in Saha regime, using either Eq.(12) or Eqs.(13) to (15) depending on if we include helium or not, over the whole time interval, spanning from $x_{\rm start} = \ln(10^{-8})$ up to $x_{\rm end} = 0$;

(ii) then we search through the Saha solution and identify when the approximation breaks down (i.e. when $X_e < 0.99$) and start solving the Peebles ODE, Eq.(2.1.1), from this point onwards, replacing the values from the Saha solution along the way;

(iii) lastly, we the reionisation contribution at late times to the Peebles solution, as described by Eq.(17), and spline the result at the end.

The hydrogen-only Saha equation, Eq. (12), reduces to a simple quadratic in X_e, which can be solved analytically. Retaining the physically meaningful positive root, we obtain

$$X_e = \frac{-A_s + A_s\sqrt{1+4/A_s}}{2},$$

where $A_s(T_b)$ is just the RHS of Eq.(12). Going far back in time, the baryon temperature will get extremely large such that $A_s \sim T_b^{3/2}$ blows up and may cause overflows. To get around this problem we Taylor expand the square root term[2] and find that $X_e = 1$ in this limit. This is an excellent approximation as long as $A_s \gg 4$. In our numerical implementation, the approximate solution is imposed when $A_s > 400$.

If however we include helium in our Universe ($Y_p \neq 0$), we must instead solve the three Saha equations, Eqs.(13) to (15). Defining the quantities

$$r_0 \equiv \frac{2}{n_b}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_0/k_B T_b},$$
$$r_1 \equiv \frac{4}{n_b}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_1/k_B T_b},$$
$$r_2 \equiv \frac{1}{n_b}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\epsilon_0/k_B T_b},$$

we can solve for the ion matter fractions u, v, and w as given by

$$u = \frac{f_e r_0}{f_e^2 + f_e r_0 + r_0 r_1},$$
$$v = \frac{r_0 r_1}{f_e^2 + f_e r_0 + r_0 r_1},$$
$$w = \frac{r_2}{f_e + r_2}.$$

Numerically, the system is solved iteratively at each time step. Beginning with an initial guess of $f_e = 1$, we solve for the ionisation fractions u, v, and w from the three Saha equations, then update f_e using Eq. (16). This cycle is repeated until convergence, defined by $|f_{e,\rm new} - f_{e,\rm old}| < 10^{-10}$, which is typically achieved within a handful of iterations. The free electron fraction then follows as $X_e = f_e/(1 - Y_p)$.

Going into the Peebles regime, we set the initial condition of Eq.(2.1.1) to be the value of X_e from the last valid Saha solution to ensure continuity in the full solution. The only thing worth noting about the numerical implementation is that $\beta^{(2)}$ is set to zero when $\epsilon_0/k_B T_b \gg 1$ to avoid numerical overflow from the exponential factor. Specifically, we set $\beta^{(2)} = 0$ when $\epsilon_0/k_B T_b > 200$.

Finally, introducing the reionisation contributions from hydrogen and helium reionisation is as simple as adding the extra tanh-terms in Eq.(17). After that is done, we spline the resulting function X_e and n_e in log-space using cubic spline, since their values span several orders of magnitude within short time frames.

2.2.2. Calculating the optical depth and visibility function

Once we have solved for the free electron fraction, and have thus also by proxy calculated the free electron number density n_e, we calculate the optical depth from its differential form

$$\tau' = \frac{\mathrm{d}\tau}{\mathrm{d}x} = -\frac{c\sigma_T n_e(x)}{H(x)},$$

imposing the intial condition $\tau(0) = 0$. This is achieved by solving the ODE backwards in time from $x_{\rm end} = 0$ to $x_{\rm start}$. We spline the result and its derivative, and calculate its second derivative by differentiating the τ' using the method built into our spline routine (Winther 2026e).

The visibility function is then calculated using Eq.(18). Splining $\tilde{g}$, we then its first two derivatives, $\tilde{g}'$ and $\tilde{g}''$, employing the same differentiating scheme for cubic splines.

[2] $\sqrt{1+x} \simeq 1 + x/2$ when $|x| \ll 1$.

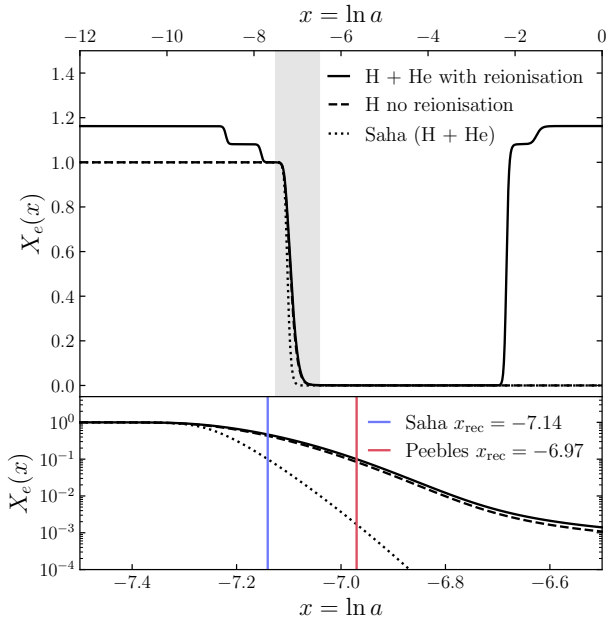


Fig. 8. The free electron fraction X_e as a function of $x = \ln a$ for three configurations: hydrogen and helium with reionisation (solid), hydrogen only without reionisation (dashed), and the pure Saha approximation with both species and without reionisation (dotted), with $Y_p = 0.245$ where applicable. The bottom panel magnifies the recombination epoch ($-7.5 \leq x \leq -6.5$, shaded grey in the top panel), highlighting the different recombination times.

2.2.3. Calculating the sound horizon at decoupling

To get the sound horizon as a function of x, we solve the ODE stated in Eq.(19) using the usual method. The initial condition is found deep in the radiation dominated era of the Universe, when $\mathcal{H}(x) \sim H_0\sqrt{\Omega_{R0}}e^{-x}$ and $R \to \infty$. The sound speed of the photon-baryon plasma is slowly varying, and approximately constant, since

$$\lim_{x\to-\infty} c_s(x) = \frac{c}{\sqrt{3}} \lim_{R\to\infty} \frac{1}{\sqrt{1+1/R}} = \frac{c}{\sqrt{3}}.$$

We then get the initial condition

$$s(x_{start}) = \frac{1}{H_0\sqrt{\Omega_{R0}}} \int_{-\infty}^{x_{\rm start}} c_s(x)\, e^x \mathrm{d}x = \frac{c_s(x_{\rm start})}{\mathcal{H}(x_{\rm start})}.$$

After splining the result, we plug in the value of x at the time of decoupling, defined as when $\tau(x_{\rm dec}) = 1$ or $\tilde{g}$ is maximal, to get the sound horizon at decoupling $r_s \equiv s(x_{\rm dec})$.

2.3. Results

2.3.1. The free electron fraction and recombination

Figure 8 displays the evolution of the free electron fraction X_e as a function of x in three different situations; the first is the fiducial model which takes into account both hydrogen and helium ionisation (with $Y_p = 0.245$) and reionisation, the second is the simplest model, corresponding to a hydrogen only Universe with no reionisation, and the third is the pure Saha solution for a hydrogen and helium Universe ($Y_p = 0.245$) with no reionisation.

The top panel shows the overall evolution of X_e. At early times the plasma is fully ionised, with helium doubly ionised ($u = 0,\ v = 1,\ w = 1$), giving

$$X_e = \frac{f_e}{1-Y_p} = \frac{1-Y_p/2}{1-Y_p} \simeq 1.162.$$

As the Universe cools and $k_BT_b \lesssim \chi_1$, helium transitions to being singly ionised ($u = 1,\ v = 0,\ w = 1$), reducing $X_e \simeq 1.081$ at around $x \sim -8.7$. A further drop occurs when $k_BT_b \lesssim \chi_0$, at which point helium becomes fully neutral and only hydrogen remains ionised ($X_e = 1$). The final and most dramatic decrease corresponds to hydrogen recombination, after which X_e freezes out at a relic plateau of $\sim 10^{-4}$. In the reionisation model, X_e rises sharply back toward the initial $X_e \simeq 1.162$ at $x \sim -2.5$ as the first stars reionise the intergalactic medium.

The hydrogen-only model without reionisation follows the same qualitative behaviour, but lacks both the early helium transitions and the late-time reionisation upturn.

The bottom panel of Figure 8 magnifies the hydrogen recombination epoch ($-7.5 \leq x \leq -6.5$) on a logarithmic scale, allowing a clearer comparison between the Saha and Peebles solutions. The pure Saha approximation predicts a significantly faster decline in X_e, continuing to decrease toward machine zero as it assumes thermal equilibrium is maintained throughout. This leads to an earlier predicted time of recombination (defined as when $X_e = 0.1$) with $x_{\rm rec} = -7.14$ compared to $x_{\rm rec} = -6.97$ from the Peebles solution. The two predictions are compared in terms of x, redshift z, and cosmic time t in Table 2.

The final freeze-out abundance of free electrons today was calculated to be $X_e(0) = 2.674\times10^{-4}$ in the fiducial model ($Y_p = 0.245$) without reionisation.

2.3.2. The optical depth and visibility function

Figure 9 shows the optical depth and its first two derivatives with respect to x for four configurations: with and without helium ($Y_p = 0.245$ or $Y_p = 0$), and with and without reionisation. Panel (a), the hydrogen-only case without reionisation, shows a smooth monotonically increasing τ going back in time, crossing $\tau = 1$ near the epoch of recombination, beyond which the Universe is optically thick ($\tau \gg 1$).

Introducing reionisation, as in panels (b) and (d), imprints a secondary feature at $x \sim -2.5$ where the sudden rise in X_e causes τ to plateau at $\tau \sim 0.1$ between the reionisation epoch and decoupling, reflecting the additional Thomson scattering experienced by CMB photons as they traverse the reionised intergalactic medium.

Including helium, as in panels (c) and (d), leaves its most visible imprint on the second derivative τ'', which develops additional small bumps at the epochs of helium recombination and reionisation. These correspond directly to the stepwise drops seen in X_e in Figure 8, where each transition in the helium ionisation state produces a brief change in the rate at which τ evolves.

Figure 10 shows the visibility function $\tilde{g}$ and its first two derivatives for the fiducial model with both hydrogen and helium. The visibility function is sharply peaked at $x_{\rm dec} = -6.973$, marking the epoch of decoupling, commonly referred to as the surface of last scattering, where the probability of a CMB photon undergoing its last Thomson scattering is maximised. The sharpness of this peak reflects the relatively brief duration of recombination: although recombination is not instantaneous, the transition from an opaque to a transparent Universe occurs over

Decoupling and recombination times ($Y_p = 0.245$)				
Event		Log scale factor x	Redshift z	Time t [Myr]
Decoupling	Peebles	-6.9728	1066.23	0.3861
	Saha	-7.1581	1283.46	0.2822
Recombination	Peebles	-6.9701	1063.36	0.3878
	Saha	-7.1404	1260.87	0.2909

Table 2. The times for decoupling and recombination for the pure Saha solution and the Peebles solution expressed in terms of x, redshift z, and cosmic time t.

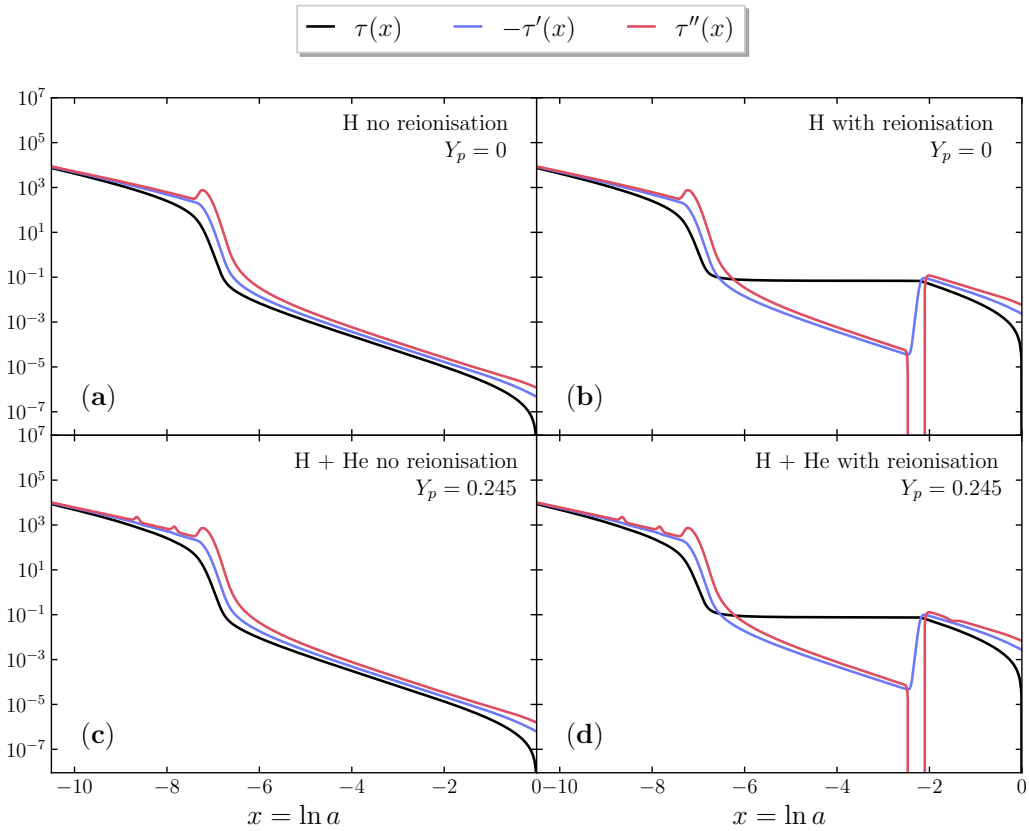


Fig. 9. Optical depth τ and its derivatives τ', τ'' as function of x for four different configurations; either with or without helium, and with or without reionisation at late times.

a narrow range of x, so the surface of last scattering is well-defined.

The small bump visible at $x \sim -2.5$ corresponds to reionisation, where the rise in X_e briefly increases the scattering rate and produces a small but non-zero probability of late-time scattering. Since $\tilde{g}$ represents the probability density that a photon was last scattered at a given x, it must satisfy the normalisation condition $\int_{-\infty}^{0} \tilde{g}(x)\,\mathrm{d}x = 1$, which our numerical implementation reproduces to machine precision, confirming the consistency of our solution. The derivatives $\tilde{g}'$ and $\tilde{g}''$ are smooth and well-behaved, and will be needed as source terms when computing CMB temperature and polarisation anisotropies. The decoupling times predicted by the Saha approximation and the Peebles equation are compared in Table 2.

2.3.3. Sound horizon at decoupling

Evaluating the sound horizon at decoupling r_s for the fiducial model, i.e. at $x_{\rm dec} = -6.9728$ (corresponding to redshift $z_{\rm dec} = 1066.23$), we find

$$r_s \equiv s(x_{\rm dec}) \simeq 146.501 \text{ Mpc}.$$

This value is in good agreement with the standard result from *Planck* (Aghanim et al. 2020), and sets a fundamental physical scale in the Universe. At decoupling there is no longer any radiative pressure to counteract the gravitational collapse of the baryons, so the acoustic oscillations in the photon-baryon plasma are frozen in place, and the sound horizon marks the maximum distance over which these oscillations could have coherently developed. This scale is directly imprinted as the characteristic angular separation of the acoustic peaks in the CMB power spec-

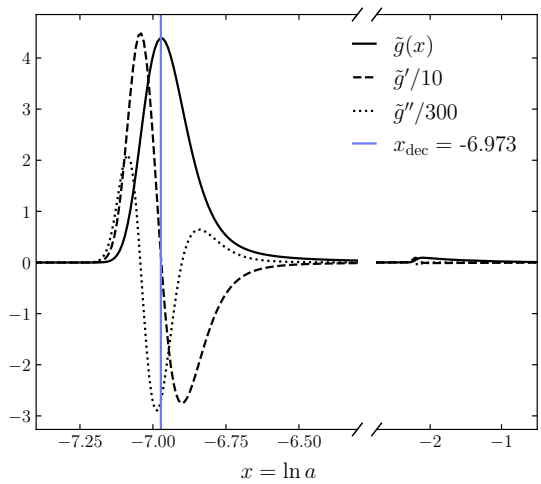


Fig. 10. Visibility function $\tilde{g}$ and its two first derivatives $\tilde{g}'$, $\tilde{g}''$ as functions of x for the Universe with both hydrogen and helium ($Y_p = 0.245$) and late-stage reionisation.

trum, and as the baryon acoustic oscillation (BAO) scale in the matter power spectrum.

Comment pure Saha decoupling/recombination and Peebles

3. Evolution of structure in the Universe: introducing perturbations

Some introduction about what it is all about.

3.1. Theory

The theory behind this milestone.

3.2. Implementation details

Something about the numerical work.

3.3. Results

4. Milestone IV

Some introduction about what it is all about.

4.1. Theory

The theory behind this milestone.

4.2. Implementation details

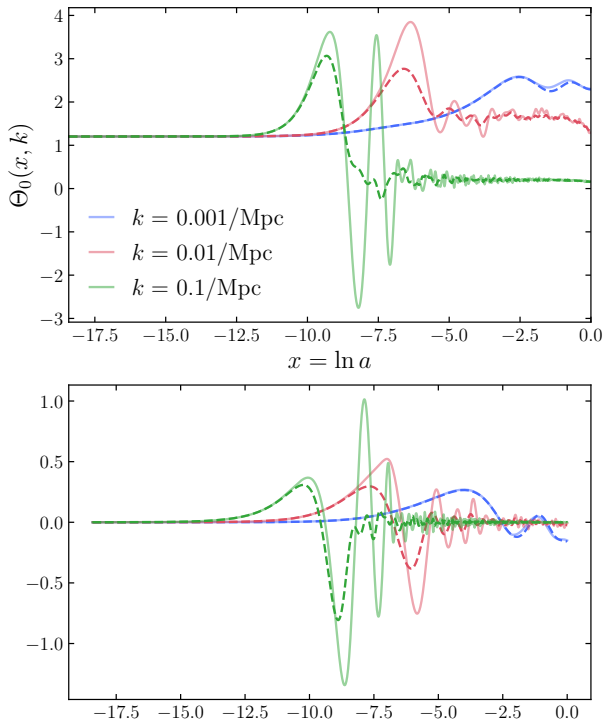
Something about the numerical work.

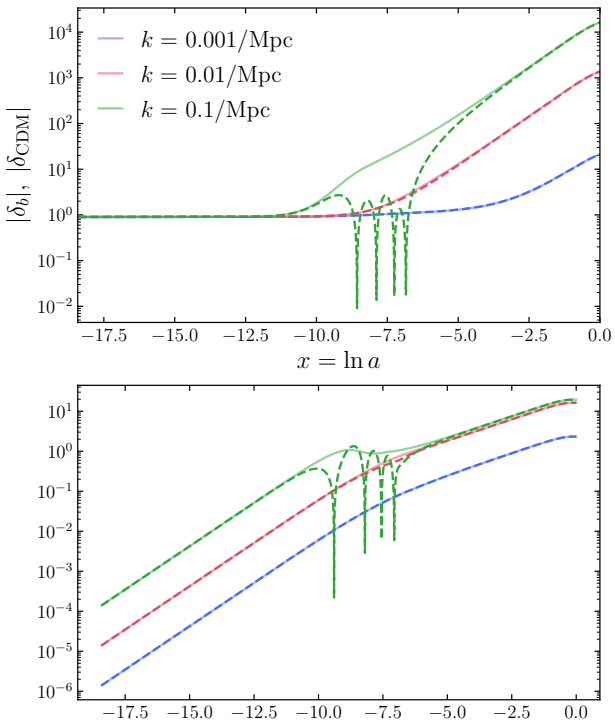
4.3. Results

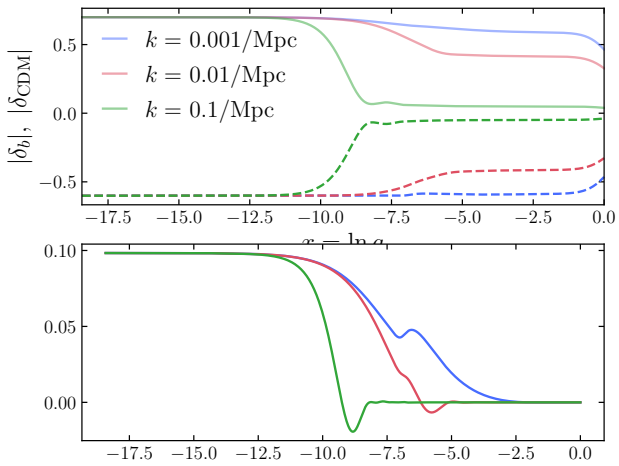
5. Conclusions

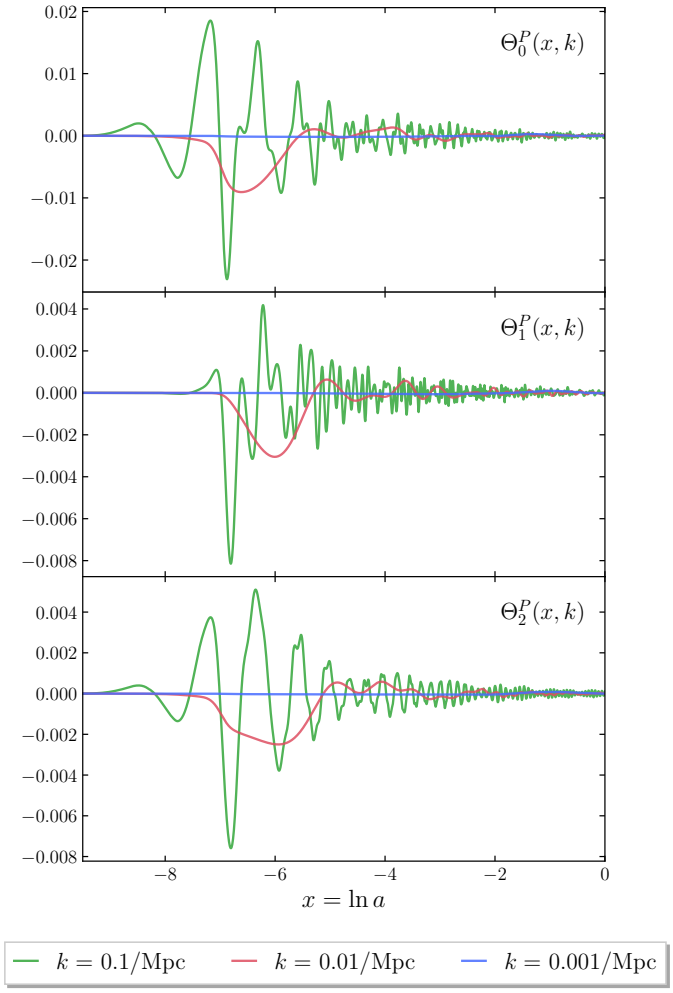
I will write a short summary and conclusion in the end.

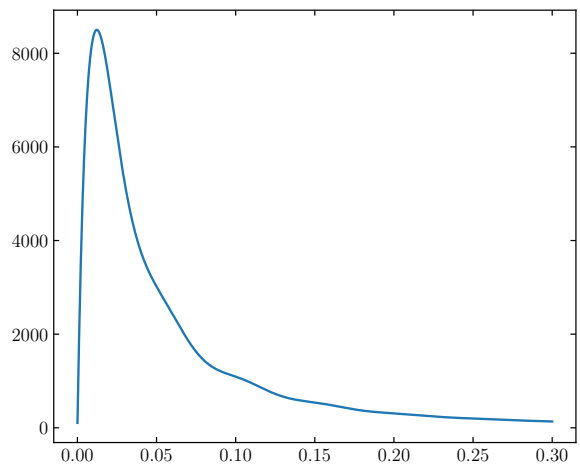
Acknowledgements. Thank you to Hans A. Winther for brilliant lectures and help with the project.











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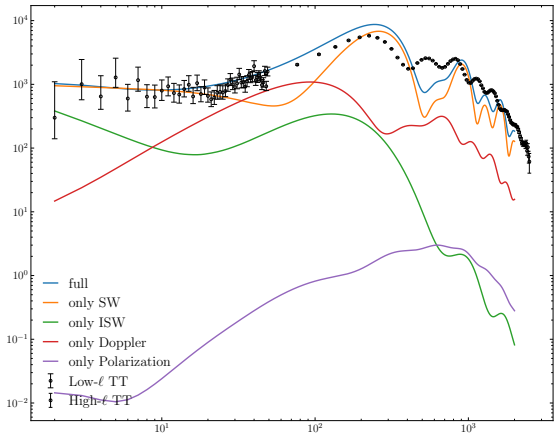
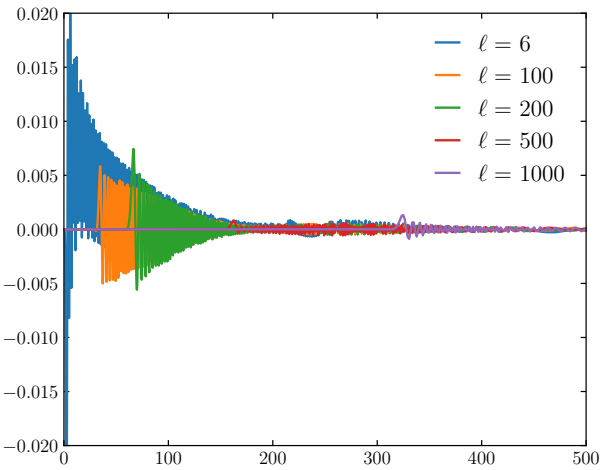

full
only SW
only ISW
only Doppler
only Polarization
Low-ℓ TT
High-ℓ TT


ℓ = 6
ℓ = 100
ℓ = 200
ℓ = 500
ℓ = 1000

Appendix A: Derivatives of the conformal Hubble function

The Hubble function is given by

$$H = H_0 \sqrt{(\Omega_{b0} + \Omega_{\rm CDM0})a^{-3} + (\Omega_{\gamma 0} + \Omega_{\nu 0})a^{-4} + \Omega_{k0}a^{-2} + \Omega_{\Lambda 0}},$$

where we can define $\Omega_{M0} \equiv \Omega_{b0} + \Omega_{\rm CDM0}$ and $\Omega_{R0} \equiv \Omega_{\gamma 0} + \Omega_{\nu 0}$. The first derivative of the conformal Hubble function $\mathcal{H} \equiv aH(a)$ with respect to $x \equiv \ln a$ is

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} = \frac{\mathrm{d}a}{\mathrm{d}x}H + a\frac{\mathrm{d}H}{\mathrm{d}x} = \mathcal{H} + a^2\frac{\mathrm{d}H}{\mathrm{d}a}.$$

Let us compute the derivative of $H(a)$:

$$\frac{\mathrm{d}H}{\mathrm{d}a} = -\frac{H_0^2}{2H}\Big(3\Omega_{M0}a^{-4} + 4\Omega_{R0}a^{-5} + 2\Omega_{k0}a^{-3}\Big).$$

This gives

$$\begin{aligned}\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} &= \mathcal{H} - a^2\frac{H_0^2}{2H}\Big(3\Omega_{M0}a^{-4} + 4\Omega_{R0}a^{-5} + 2\Omega_{k0}a^{-3}\Big)\\ &= \mathcal{H}\Big[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\Big].\end{aligned}$$

Here the Ω_i's refer to the density parameters as functions of scale factor, i.e. $\Omega_i(a)$, which should be evaluated at $a = e^x$.

The second derivative is a bit more cumbersome to calculate; from the product rule it follows that

$$\begin{aligned}\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} &= \mathcal{H}\Big[1 - \frac{3}{2}\Omega_m - 2\Omega_r - \Omega_K\Big]^2\\ &\quad + \mathcal{H}\frac{\mathrm{d}}{\mathrm{d}x}\Big[1 - \frac{3}{2}\Omega_m - 2\Omega_r - \Omega_K\Big].\end{aligned}$$

Let B be the expression inside the big bracket. We thus need $\mathrm{d}B/\mathrm{d}x$, which is most easily calculated by factoring out $H_0^2/H(a)^2$ from the last three terms of B;

$$\begin{aligned}\frac{\mathrm{d}B}{\mathrm{d}x} &= -\frac{a}{2}\frac{\mathrm{d}}{\mathrm{d}a}\left(\frac{H_0^2}{H^2}\right)\Big(3\Omega_{M0}a^{-3} + 4\Omega_{R0}a^{-4} + 2\Omega_{K0}a^{-2}\Big)\\ &\quad - \frac{a}{2}\frac{H_0^2}{H^2}\frac{\mathrm{d}}{\mathrm{d}a}\Big(3\Omega_{M0}a^{-3} + 4\Omega_{R0}a^{-4} + 2\Omega_{K0}a^{-2}\Big)\\ &= -\frac{1}{2}\Big(3\Omega_M + 4\Omega_R + 2\Omega_K\Big)^2 + \frac{1}{2}\Big(9\Omega_M + 16\Omega_R + 4\Omega_K\Big).\end{aligned}$$

From this we end up with second derivative

$$\begin{aligned}\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} &= \mathcal{H}\Bigg\{\Big[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\Big]^2\\ &\quad - \frac{1}{2}\Big(3\Omega_M + 4\Omega_R + 2\Omega_K\Big)^2 + \frac{1}{2}\Big(9\Omega_M + 16\Omega_R + 4\Omega_K\Big)\Bigg\}\Bigg|_{a=e^x}.\end{aligned}$$

Appendix B: Logarithmic scale factor at different cosmological events

Matter-radiation equality happens when $\Omega_M(a_{\rm eq}^{MR}) = \Omega_R(a_{\rm eq}^{MR})$. Using the expressions given in Eq.(5), we find

$$\frac{\Omega_{M0}}{(a_{\rm eq}^{MR})^3} = \frac{\Omega_{R0}}{(a_{\rm eq}^{MR})^4} \;\Rightarrow\; x_{\rm eq}^{MR} \equiv \ln a_{\rm eq}^{MR} = \ln\left(\frac{\Omega_{R0}}{\Omega_{M0}}\right).$$

In a similar way, we derive that the matter-dark energy equality occurs for

$$\frac{\Omega_{M0}}{(a_{\rm eq}^{\Lambda M})^3} = \Omega_{\Lambda 0} \;\Rightarrow\; x_{\rm eq}^{\Lambda M} \equiv \ln a_{\rm eq}^{\Lambda M} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{\Omega_{\Lambda 0}}\right).$$

The onset of cosmic acceleration is defined to be when $\ddot{a} = 0$. From the second Friedmann equation, Eq.(3), we get that this occurs whenever

$$\sum_i \rho_i(a_{\rm acc})(1 + 3w_i) = 0.$$

The relevant equations of state are $w_M = 0$, $w_R = 1/3$ and $w_\Lambda = -1$. We neglect the curvature parameter when working with fiducial cosmological parameter, and thus the condition for cosmic acceleration can be recast into

$$\begin{aligned}&\Omega_M + 2\Omega_R - 2\Omega_\Lambda = 0\\ \Rightarrow\quad &\frac{\Omega_{M0}}{(a_{\rm acc})^3} + \frac{2\Omega_{R0}}{(a_{\rm acc})^4} - 2\Omega_{\Lambda 0} = 0.\end{aligned}$$

At late times, the radiation density parameter can be neglected, so we end up with

$$x_{\rm acc} \equiv \ln a_{\rm acc} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{2\Omega_{\Lambda 0}}\right).$$